

RAMNATH RAMACHANDRAN

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<https://www.linkedin.com/in/ramnath94/> | [GitHub Projects Repository](#)

- **Analytics professional and a Machine learning enthusiast with 2 years of relevant academic and practical experience and 2+ years of professional experience.**
- **Knowledgeable in fields like Machine Learning, Software Development, Statistics, Data Mining, and Proficient in tools like Python, R, SQL, Tableau, and Java.**

EDUCATION

Master of Science (M.S.) - Business Analytics, the University of Illinois at Chicago (GPA: 3.88) December 2020
Bachelor of Engineering (B.E.) - Computer Science, Anna University, India (GPA: 7.5/10) May 2016

COURSEWORK

Graduate: Business Data mining, Statistics for Management, Machine learning and Advanced Statistics, Analytics for Big Data, Business Data Visualization, Health Information Analytics, Machine learning Deployment, Time Series & Forecasting

Undergraduate: Data structures, Analysis of Algorithms, Java and Internet Programming

CERTIFICATIONS

[Deep Learning Specialization \(Coursera\)](#) - [Tableau Desktop Specialist](#) - [Applied Statistical Modeling for Data Analysis in R](#)

SKILLS

Core Competencies: Machine learning, Predictive Analytics, Data Visualization, Data Wrangling, Statistical Analysis, Deep Learning, Text mining and analytics, Healthcare Analytics

Programming/Scripting: Python, R, SQL, Java, C, C++, JavaScript, MATLAB, HTML5, CSS

Analytics/Visualization: Tableau, Excel, SPSS, Spark

Databases: MS-SQL Server, MySQL, PostgreSQL

Tools/Framework: Git, Hadoop, Jupyter notebook, Rstudio, Tableau Desktop, Selenium, Docker, AWS EC2, S3, Sagemaker, Mule Anypoint, Jenkins, Microsoft Office, Linux Scripting, MLflow, Putty

Analytics Packages: pandas, numpy, Scikit-learn, PySpark, nltk, Keras, Tensorflow, Pytorch, plotly, matplotlib, dplyr, tidyverse, caret, ggplot2, data.table, forecast, mlflow

EXPERIENCE

Summer Research - The University of Illinois at Chicago | NLP, Deep learning, nltk, Pytorch 05/2020 - Present

- Machine learning for Data processing: Working towards processing the online Thesaurus data of Merriam webster into operable data using intelligent computational models.
- Performed pre-processing on the web-scraped Thesaurus of words, meanings, and examples by chunking and tagging the data to form a pattern using nltk tags.
- Researching about efficient Deep NLP techniques to experiment on the pre-processed Dictionary data to achieve a well-structured Merriam webster Thesaurus.

Software Engineer - Accenture solutions, Chennai, India | Java, Mulesoft, Tableau 06/2018 – 06/2019

- Performed data analysis on the performance of the eRPI application in the production environment with the help of the Tableau software tool.
- Successfully implemented the backend support system for Application Request Management (ARM) ticket processing system for the client portal using Java web framework and JDBC.

Associate Software Engineer - Accenture solutions, Chennai, India | Java, Mulesoft, Tableau 08/2016 – 05/2018

- Successfully designed and built the Transformation and Routing module of the eRPI(Embedded Request Processor Integrator) a message-oriented middleware system built to provide efficient integration solutions to online banking operations.
- Analyzed and organized client needs throughout by collaborating with the help of the Agile methodology and Continuous Integration tools like Jenkins, Bitbucket.

PROJECTS

08/2019 - Present

Text Analytics on Pediatricians | Text Analytics, NLP, Regression, Web Scraping, Python, Excel, Selenium

- Successfully incorporated Predictive analytics and text mining to examine how pediatricians around the state Illinois are reviewed or rated online on different aspects of their services on healthcare websites like RateMDs and Healthgrades.
- Performed sentiment analysis on web-scraped text data consisting of online reviews of 1000 pediatricians followed by Semi-supervised topic modeling to calculate aspect-based scores for each pediatrician.
- Built a regression model with an overall rating as a target against the calculated aspect scores and various other review characteristics and doctor demographics to find the key factors affecting the star ratings.

Data Analytics on Live music events in the UK | ETL process, Analytics & Visualization, Python, PySpark, SQL, Tableau

- Extracted data on live music performances taken between March 2019-Feb 2020 from Spotify and MusicBrainz consists of 1 million records to recommend live music event dates, time, and venue to the public through analytics.
- Performed data cleaning on the extracted data and persisted in the PostgreSQL database for further analytical processing.
- Visually analyzed the loaded final data using Tableau to derive inferences on artists' popularity based on their event popularity, a perfect venue to host, date-wise, and hour wise event counts and stream counts.

Network Traffic Intrusion Prediction | Big Data, Predictive Analytics | Databricks, Spark ML, Python, Mlflow, AWS

- Successfully created Spark ML pipeline models to predict the potential network intrusion given U.S Airforce TCP LAN dump data consisting of 5 million records stored in the AWS S3 storage.
- Serialized the best performing model with 98% Accuracy and F1- score into a bundle with the help of Mlflow's mleap format and exported the model for Predictions.

Real-time Car accident Severity prediction | Predictive Analytics, ML Deployment | AWS- S3, Sagemaker, Docker, Python

- Incorporated predictive analytics models to predict the severity of Car accidents across Illinois given the US real time car accident data with Event time, geographic location and weather-related information as features.
- Deployed the best performing model Randomforests with 94% F1-score into the Amazon Sagemaker with the help of Mlflow python API and containerized with the help of Docker in the AWS EC2 instance.